

Deep learning-based direct and indirect potential evapotranspiration prediction, a case of the Nakdong River basin, South Korea

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ABSTRACT

Accurate estimation of the potential evapotranspiration (PET) is crucial for hydrological modeling and water resource management. This study develops and compares two deep learning (DL) models, namely an indirect graph convolutional recurrent network (GCRN) and a direct GraphWaveNet model, to predict daily PET in the Nakdong River basin, South Korea. Reference PET was calculated from 52 years (1973–2024) of meteorological data using the FAO-56 Penman-Monteith (PM) method. The indirect framework predicts variables (net radiation, vapor pressure, and vapor-pressure deficit) prior to PET recomposition, whereas the direct framework predicts PET with reduced inputs on a data-scarce basis. Quantitative evaluation across 13 stations demonstrates that GraphWaveNet outperforms GCRN, with RMSE ranging from 0.52 to 0.68 mm/day and MAE values ranging from 0.40 to 0.50 mm/day, which are 50–65% lower than GCRN's (RMSE values ranging from 1.3 to 1.6 mm/day). The direct model also yielded a higher R^2 (0.85–0.92) than GCRN (0.14–0.48) and had a slight bias (<0.08). GraphWaveNet captured seasonal and short-term variations while maintaining variance; however, the indirect GCRN was underestimated under high-PET conditions due to errors propagated through the reconstructed PM components. Findings confirm that adaptive spatiotemporal graph-based DL has a robust, precise, and physically coherent approach to PET estimation.

1. Introduction

Potential evapotranspiration (PET) is a key component of the water cycle, playing a key role in hydrological modeling, agricultural water management, and climate-impact studies (Vörösmarty et al., 1998; Xiong et al., 2024). PET is a key element in applications as diverse as irrigation scheduling and drought or land-surface modeling, because accurate estimation is imperative when water availability is assumed to be ideal (Okkan et al., 2024). The FAO-56 Penman-Monteith (PM) formulation has been widely accepted as the most physically sound and transportable method for determining PET across a range of climatic regimes, due to its physically explanatory form and excellent performance in diverse climatic conditions (Allen et al., 1998). Nonetheless, the PM method requires high-quality input data on meteorological conditions, which are often incomplete or lack adequate spatial or temporal resolution in many environmental monitoring systems (Allen et al., 2011; Almorox et al., 2018). These limitations have sparked growing interest in data-driven and deep learning (DL) approaches that

can learn non-linear relationships between climatic drivers and predict PET under both data-intensive and data-scarce conditions (Sun et al., 2024; Yan et al., 2023).

DL-based estimation of climatological variables has advanced significantly, and models such as multilayer perceptron (MLPs) (Samii et al., 2023), convolutional neural networks (CNNs) (García-Pedrero et al., 2017), and recurrent neural networks (RNNs) (Ali et al., 2025) have proven proficient in capturing non-linear temporal dependencies and enhancing hydrometeorological forecasts across various areas. DL models have effectively been applied to predict daily or hourly PET using only a few meteorological variables, particularly when no radiation data are available or physical models are computationally intensive (Ferreira et al., 2022). The RNN-based models, such as long short-term memory (LSTM) networks and gated recurrent units (GRUs), have demonstrated good performance in modeling temporal dynamics and seasonal variability (Hilmi et al., 2022).

In contrast, CNN-based models are effective in extracting local feature patterns from time series (Waqas and Wangwongchai, 2025).

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However, one significant drawback of traditional DL structures is that they assume the input variables are independent features (Waqas et al., 2023). This assumption contradicts the interdependence of environmental processes, particularly those of PET, in which temperature, humidity, vapor pressure deficit, radiation, and wind velocity interact in well-established physical relationships (Nalevanková et al., 2020).

New directions in graph neural networks (GNNs) offer a potential solution to these shortcomings by modeling the structure of relationships among variables using graph representations (Waikhom and Patgiri, 2023; Waqas et al., 2026; Wu et al., 2020). GNNs are designed to handle graph-structured data, where a relational dependency between nodes can significantly impact learning via graph convolution or message passing (Wu et al., 2020). Structural GNNs are defined by modeling variables as nodes and their relationships as edges, allowing the learning of node-level representations to depend on the graph's topology and interaction intensities (Telyatnikov, 2025; Waqas et al., 2026; Zhao et al., 2021).

This model applies to hydroclimatic modeling, as variables of the PET are physically interrelated. Vapor pressure depends on temperature and humidity, and the net radiation time is controlled by the length of the sunshine period, via known atmospheric relations (Gyamfi et al.; Jia et al., 2023). GNNs enable the incorporation of domain knowledge and observational relationships into a graph format, thereby increasing model interpretability, stability, and prediction accuracy (Zhao et al., 2024). Although GNNs have become effective at modeling relational environmental systems, their use for predicting evapotranspiration remains scarce (Gyamfi et al.; Jia et al., 2023; Zhao et al., 2024). Current studies use conventional DL frameworks or regression models based on empirical evidence, and comparatively little has been done to investigate the applicability of graph-convolutional or graph-recurrent networks to PET estimation.

Past research has primarily focused on direct prediction, i.e., predicting PET based solely on existing climate variables, without considering hybrid or indirect approaches that explicitly account for physical constraints (Slater et al., 2023; Wi and Steinschneider, 2024). Even simple data-driven models can be highly predictive; however, they are not based on physical processes and are unstable in the presence of climate variability and extremes (Knüsel and Baumberger, 2020). The combination of GNNs and process-based formulations, including the PM equation, presents an opportunity to narrow this gap and leverage the advantages of both physical and data-driven approaches.

Another research gap arises from the need to test PET prediction models under both data-rich and data-scarce conditions (Li et al., 2021). In most applications, radiation or humidity measurements can be lost, sensors can be faulty, or long-term historical measurements can include gaps (Rana and Katerji, 2000). Most literature is based on models trained under ideal data conditions, with small consideration for situations where significant inputs are missing (Ferreira et al., 2022; García-Pedrero et al., 2017; Hilmi et al., 2022). It poses an essential problem for operational PET estimation systems and climate-impact models, which must remain robust despite limited data availability. The Nakdong River basin in South Korea presents a valuable case study for evaluating the PET estimation models. It provides a valuable research opportunity for PET modeling (Lee et al., 2009; Noh et al., 2016).

Therefore, there is a need to adopt a systematic methodology that integrates long-term meteorological records, physically based PET calculations, and a state-of-the-art GNN. This research fills gaps by creating two complementary PET prediction pathways: first, an indirect approach using a graph-convolutional recurrent network (GCRN) to predict the physical components of PET before decomposition; and second, a direct approach using a GraphWaveNet architecture to predict PET based on meteorological variables and to learn adaptive spatio-temporal dependence. The use of graph representations to encode physical and data-driven relationships in the study improves environmental prediction modeling in a variety of ways: (i) it shows that GNNs

can capture complex non-linear interactions among meteorological variables, (ii) it evaluates the performance of models under condition of variable availability to simulate data-sparse environments, and (iii) it provides a rigorous evaluation of interpretability using feature-attribution methods, thus increasing the physical plausibility of the DL-based PET estimation.

2. Materials and methods

2.1. Study area

The Nakdong river basin is the largest watershed in southeastern South Korea, which covers approximately 25% of the national land area (Kang et al., 2020; Waqas and Kim, 2025) (Fig. 1). The basin has an area of roughly 23,400–23,600 km² and includes the 510–525 km long Nakdong River, which runs through large provinces and metropolitan regions and empties into the Korea Strait (Kang et al., 2020; Seo et al., 2021). The basin area comprises 22–33 sub-basins and 195–266 unit catchments with varied land use, including large forest cover (68%), agricultural lands, and urbanized regions (Kang et al., 2020; Seo et al., 2021). Hot and humid summers and cold, dry winters typify the climate, with an annual rainfall of about 1200 mm, of which 60% occurs during the monsoon season between June and September (Lee et al., 2020). Several dams and weirs play a significant role in altering hydrological conditions, flow regimes, and water quality dynamics, thereby influencing spatial heterogeneity that can be used in modeling evapotranspiration.

2.2. Data sources and pre-processing

The long-term observational dataset (1973 to 2024) includes daily meteorological variables, including maximum and minimum temperatures ($T_{\min}$ and $T_{\max}$), relative humidity (R_H), sunshine duration (N), and wind speed (U_2). The variables represent the default input set of reference PET and are widely applied in hydroclimatic and agrometeorological applications. The raw data underwent multi-stage quality control to ensure completeness and temporal stability before analysis. The missing data were detected and imputed using DL-based imputation techniques that were suitably tuned to preserve intra-seasonal climatological variability. Interquartile-range-based outlier identification was used to filter potentially erroneous observations, which were then further evaluated against long-term statistical distributions (Table 1).

2.3. PET estimation using the FAO-56 Penman–Monteith method

The daily PET was estimated using the standardized FAO-56 PM formulation (Allen et al., 1998). In the PM model, R_n was used to determine net radiation, and the heat flux to the soil was considered insignificant at the daily scale. All variables were estimated according to the FAO-56 (Allen et al., 2011, 1998). A 52-year PET series that emerged from this served as the ground truth for training the DL model. The method accounts for the effects of net radiation (R_n), air temperature $T_{\max}$ and $T_{\min}$, R_H , and u_2 on evapotranspiration processes. The FAO-56 PM equation is expressed as:

$$PET = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T+273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \quad (1)$$

where:

PET is the potential evapotranspiration (mm day⁻¹), Δ is the slope of the saturation vapor pressure curve (kPa °C⁻¹), R_n is the net radiation at the crop surface (MJ m⁻² day⁻¹), and G is the soil heat flux (MJ m⁻² day⁻¹). γ is the psychrometric constant (kPa °C⁻¹), T is the mean daily air temperature (°C), u_2 is the wind speed at 2 m height (m s⁻¹), e_s is the saturation vapor pressure (kPa), and e_a is the actual vapor pressure

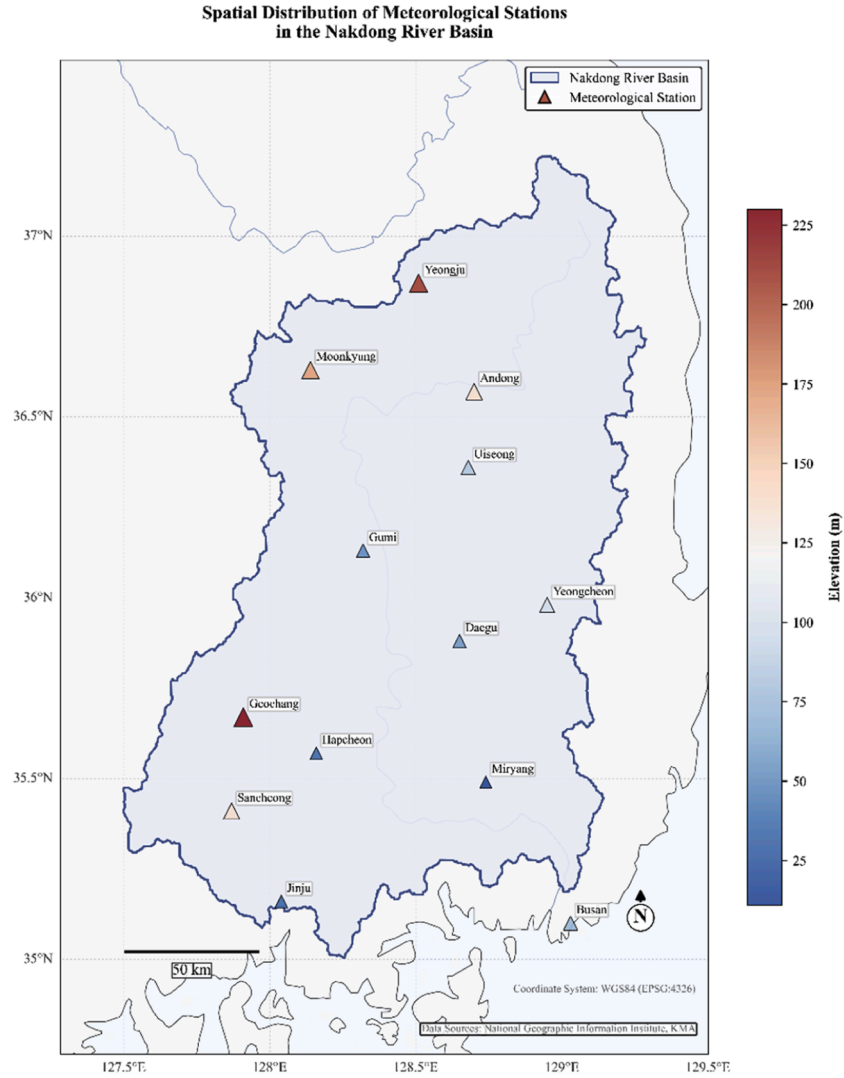


Fig. 1. Distribution of meteorological stations over the Nakdong River basin.

(kPa).

2.4. DL frameworks

The DL frameworks were designed to evaluate and enhance daily PET estimation with two supplementary predictive methods. Indirect approach is a procedure whereby the DL models are trained to predict the FAO 56-PM PET using daily meteorological variables; in other words, they acquire the non-linear association between climate drivers and the desired PET, as illustrated in Fig. 2. Conversely, the direct method forecasts PET using fewer climatic variables, excluding FAO 56-PM outputs, to simulate data-sparse conditions and assess the ability of DL models to predict PET with limited inputs.

2.4.1. GNN architecture for daily PET prediction

The primary methodological innovation of the current study is the GNN architecture, which is designed to predict daily PET. Unlike traditional DL models, which assume input variables are independent features, GNNs use graph-based representations to capture intricate relational patterns among variables (Wu et al., 2020). In this work, meteorological variables were used as nodes of a graph. The resultant variable-interaction graph, built using the hybrid adjacency matrix as outlined in Section 3.4, captures the strength of interactions among the climate variables and serves as the foundation of the station interaction

network, as shown in Fig. 3.

Important relationships were captured as edges: temperature-humidity to vapor pressure deficit and sunshine duration-radiation to net radiation. The FAO-56 PM equation and data correlation were used to determine the graph structure, ensuring physically realistic dependencies. The GNN architecture was further developed with recurrent temporal modules, including GRU-based sequence encoders, to capture temporal evolution within a spatiotemporal graph neural network, as in graph convolutional recurrent networks (Seo et al., 2018). The model output layer directly forecasted daily PET, and training was performed using the same loss functions and optimization methods as for the neural baselines.

Indirect GNNs use predicted intermediate physical components (e.g., net radiation, R_n , and vapor pressure) as input into the PM formula; direct GNNs predict daily PET directly from input node features. The central hypothesis is that the interdependencies between variables, represented as a graph, encode predictive ability and physical stability more effectively than traditional (non-graph) deep models (Allen et al., 1998).

(i) Indirect GCRN for daily PET estimation

Daily meteorological dataset spans T days and contains raw variables $\{X_t\}_{t=1}^T$ where a single-day feature vector $X_t \in \mathbb{R}^V$ collects variables

Table 1
Statistics of input variables across stations in the basin.

T_{min} (°C)				T_{max} (°C)			WS (m/s)			R_H (%)			Sunshine (hours)		
Station ID	Mean	Std	CV (%)	Mean	Std	CV (%)	Mean	Std	CV (%)	Mean	Std	CV (%)	Mean	Std	CV (%)
136	6.69	10.60	158.40	18.29	10.11	55.26	1.62	0.89	54.73	67.45	15.01	22.25	6.35	3.71	58.39
143	9.53	9.78	102.64	19.63	9.78	49.80	2.57	1.21	46.90	61.79	15.46	25.02	6.33	3.84	60.63
159	11.52	8.75	75.92	19.06	7.87	41.31	3.62	1.43	39.64	64.58	17.72	27.44	6.46	4.00	61.89
192	7.94	10.31	129.80	19.65	8.87	45.17	1.49	0.88	58.90	69.67	14.37	20.62	6.23	3.78	60.60
272	5.94	10.44	175.58	17.57	10.33	58.75	2.21	1.60	72.57	66.97	14.82	22.13	6.73	3.72	55.30
273	6.86	10.06	146.69	17.90	10.13	56.57	1.63	0.96	59.22	65.13	16.44	25.24	6.48	3.90	60.18
278	5.19	11.21	215.87	19.01	10.07	52.94	1.13	0.74	65.61	69.36	12.50	18.03	6.21	3.69	59.45
279	7.55	10.17	134.71	18.90	9.96	52.68	1.58	1.10	69.71	66.74	14.64	21.93	6.21	3.68	59.26
281	7.06	10.18	144.20	19.04	9.65	50.66	1.73	1.03	59.36	65.52	14.45	22.05	6.20	3.60	58.10
284	6.09	10.32	169.36	18.67	9.53	51.07	1.33	0.80	59.69	70.04	13.09	18.68	6.57	3.60	54.86
285	7.60	10.29	135.27	19.83	9.28	46.78	1.20	0.69	57.75	67.95	13.70	20.16	6.44	3.71	57.61
288	7.84	10.30	131.39	20.08	9.15	45.56	1.38	0.76	55.19	66.89	14.51	21.69	6.42	3.70	57.66
289	7.80	9.77	125.28	19.35	9.34	48.25	1.52	1.16	76.33	66.95	14.95	22.34	6.20	3.66	59.00

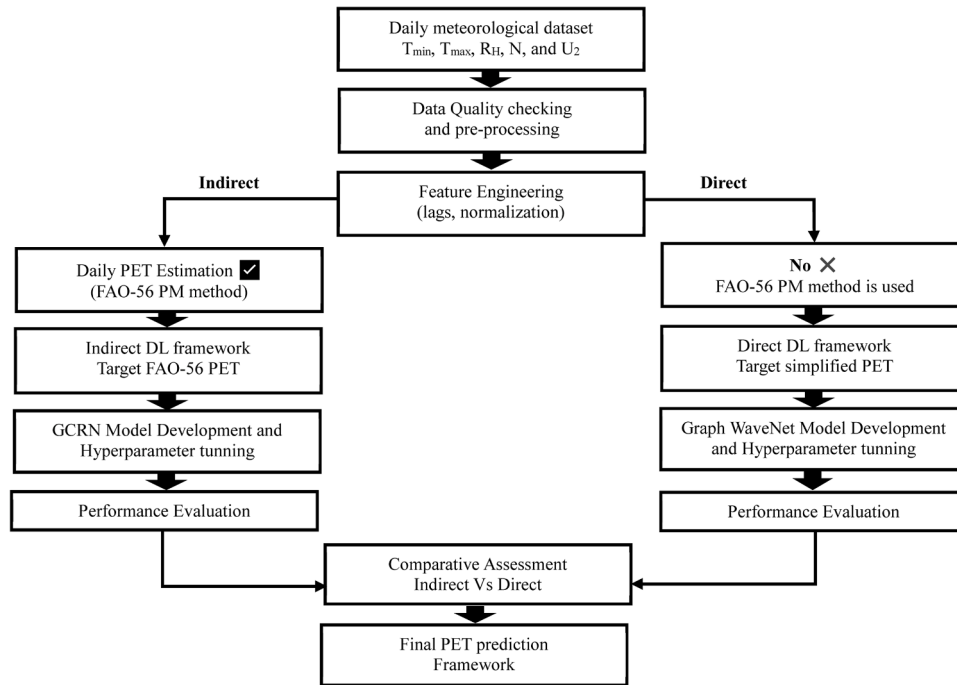


Fig. 2. Methodological framework of the study.

such as T_{max} , T_{min} , T_{mean} , R_H , wind speed at 2 m (U_2), N , and any derived diagnostics.

This model forms overlapping input sequences (windows) of length L : for the target day t , the input sequence is.

$$X_{t-L+1:t} = [X_{t-L+1}, \dots, X_t].$$

Node-wise notation: model treats each *variable* as a node in a variable-interaction graph $G = (V, E)$ with $|V| = N$ nodes (e.g., $N \approx$ number of unique variables/derived components). The graph adjacency is $A \in \mathbb{R}^{N \times N}$ (see Graph Construction Fig. 3).

In this study, nodes correspond to meteorological variables and physically derived components involved in the FAO-56 PM evapotranspiration formulation. Specifically, the node set includes T_{max} , T_{min} , T_{mean} , R_H , wind speed at 2 m (U_2), N , R_s , R_n , e_s , e_a , VPD, and the slope of the saturation vapor pressure curve (Δ). Each node therefore represents a climate variable or a derived diagnostic influencing evapotranspiration processes.

PET is denoted $E_{To}^{PM}(t)$. Moreover, computed daily, as described by Allen et al. (1998). For the indirect model, the model predicts a set of intermediate physical components $C = \{c_1, \dots, c_m\}$ (e.g., R_n , a

e_a , VPD), and then recompute $\hat{E}_{To}^{PM}(t)$ by substituting predicted components into the PM equation.

All training uses the training/validation/test chronological split (train 70%, val 15%, test 15%) to avoid temporal leakage. Feature engineering involves converting N to R_s using Ångström–Prescott equation (Angstrom, 1924); R_a and daylight fraction by day of year (Allen et al., 1998). Furthermore, computed derived variables per day: $e_s(T)$, e_a (from R_H), slope of the saturation vapor pressure curve $\Delta(T)$, and $VPD = e_s - e_a$.

These derived items are included as nodes. Standardizing features involves computing the mean and standard deviation on the training set and applying a z-score transform to all splits. For GNNs, this study also linearly embeds each raw variable to an initial node embedding. $H_{ti}^{(0)} \in \mathbb{R}^d$ via a shared linear layer per node type.

(a) Graph construction and variable-interaction modeling

Construct a hybrid adjacency $A = \alpha A^{phys} + (1 - \alpha) A^{data}$ with $\alpha \in [0, 1]$. A^{phys} encodes known FAO-56 relations (e.g., $sunshine \rightarrow R_n$; T and $R_H \rightarrow e_a$; R_n and VPD interact in the PM numerator) as binary or

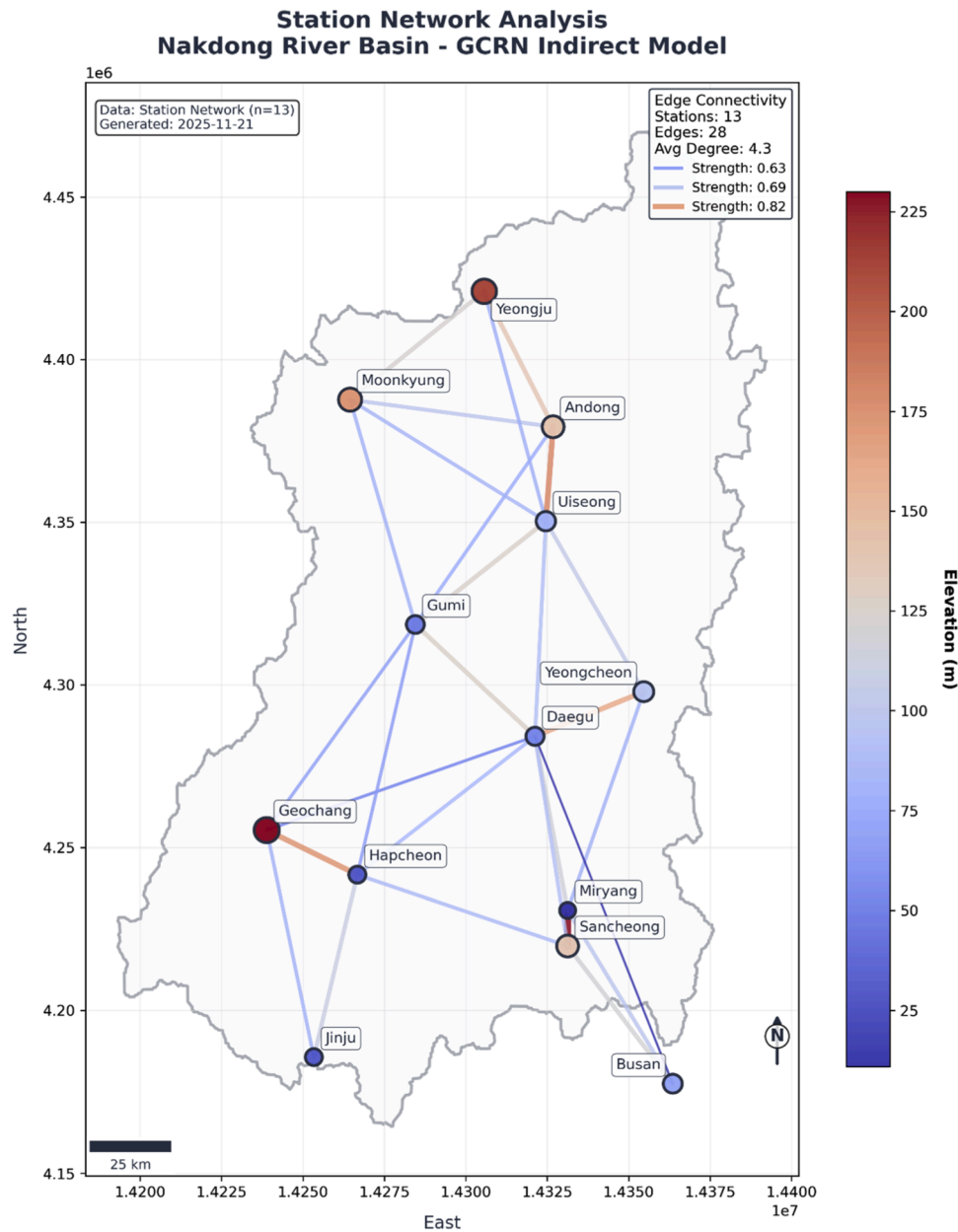


Fig. 3. Station network analysis by the GCRN indirect model.

scaled links.

The parameter α controls the relative contribution of physics-based relationships and data-driven correlations in the final graph structure. In this study, α was treated as a hyperparameter and selected through validation experiments within the range [0.3–0.7]. The optimal value ($\alpha = 0.5$) provided the best validation performance, balancing physically interpretable structure with data-adaptive relationships.

A^{data} is computed from training data using partial correlation or mutual information thresholding, then symmetrized. Alternatively, permit adaptive adjacency learning later (GraphWaveNet) by initializing with A and allowing low-rank node-embedding-based correction (Wu et al., 2019).

Edges are interactions between variables, whether statistical or physically obtained, and are represented by weighted connections in the adjacency matrix. The undirected weighted graph, where the weight of the edge measures the strength of the interaction between variables.

Two complementary adjacency matrices are defined: Physical adjacency (A^{phys}) encodes known dependencies derived from the FAO-56 PM

equation. These connections are represented as binary links indicating physically established relationships (e.g., sunshine duration $\rightarrow$ solar radiation R_s ; temperature and relative humidity $\rightarrow$ actual vapor pressure e_a ; radiation and VPD contribute jointly to PET). Data-driven adjacency (A^{data}) captures empirical relationships learned from the training dataset. It is constructed using partial correlation and mutual information analyses, in which statistically significant relationships exceeding a predefined threshold are retained as weighted edges.

The resulting adjacency matrix is symmetrized to produce an undirected interaction graph and normalized as Normalize adjacency as $\tilde{A} = D^{-\frac{1}{2}}(A + I)D^{-\frac{1}{2}}$ where D is the degree matrix.

(b) Indirect GCRN

Predict intermediate physical components $\hat{C}_t = \{\hat{R}_n, \hat{e}_a, \dots\}$ for day t using the previous $L = 14$ days of node features, then recompute $\hat{E}_{To}^{PM}(t)$ and evaluate PM-consistency.

Input sequence length $L = 14$ days. Per time step $s \in \{t-L+1, \dots, t\}$ form node features $H_s^{(0)} \in \mathbb{R}^{N \times d}$ (linear embed). The temporal stack is:

GraphConv (diffusion conv $K=2$) $\rightarrow$ Graph-GRU layer (hidden 128) $\times 2 \rightarrow$ Dense node readout $\rightarrow$ Predicted components $\rightarrow$ PM recomposition.

For a single time, slice s , diffusion graph convolution with K hops produce:

$$\mathcal{Z}(H_s) = \sum_{k=0}^{K-1} ((D_o^{-1}A)^k H_s W_k^{(f)} + (D_o^{-1}A^\top)^k H_s W_k^{(b)}), \quad (2)$$

where $W_k^{(\cdot)}$ are trainable filters, D_o , D_i are out/in degree matrices, and $K=2$ in our design.

(c) Graph-GRU layer (graph-enhanced gated recurrence)

Replace dense linear transforms in GRU with graph conv operator $\mathcal{Z}(\cdot)$. For time step s :

$$r_s = \sigma(\mathcal{Z}(X_s)W_r + \mathcal{Z}(H_{s-1})U_r + b_r) \quad (3)$$

$$z_s = \sigma(\mathcal{Z}(X_s)W_z + \mathcal{Z}(H_{s-1})U_z + b_z) \quad (4)$$

$$\tilde{H}_s = \tanh(\mathcal{Z}(X_s)W_h + \mathcal{Z}(r_s \odot H_{s-1})U_h + b_h) \quad (5)$$

$$H_s = z_s \odot H_{s-1} + (1 - z_s) \odot \tilde{H}_s \quad (6)$$

Hidden dimension per node: 128. Two stacked Graph-GRU layers process the sequence; the final hidden state for day t is $H_t^{(L)} \in \mathbb{R}^{N \times 128}$.

(d) Dense readout per node

Apply a small MLP per node (shared parameters across nodes or node-specific heads) to map node hidden vectors to predicted scalar components:

$$\hat{C}_{t,i} = \text{MLP}_i(H_{t,i}^{(L)}) \quad \forall i \in C \quad (7)$$

Aggregate predicted components $\hat{C}_t$ and compute recomposed $\hat{E}_{T_0}^{PM}(t) = PM\hat{C}_t$ by plugging predicted R_n , e_a , Δ , etc. into the FAO-56 equation.

(e) Loss function

Total loss for a sample day t :

$$\mathcal{L}_t = \mathcal{L}_{comp} \sum_{c \in C} \text{MSE}(c_t, \hat{c}_t) + \lambda_{PM} \text{MSE}(E_{T_0}^{PM}(t), \hat{E}_{T_0}^{PM}(t)), \quad (8)$$

with hyperparameters λ_{comp} , λ_{PM} controlling component vs PM fidelity (default 1.0 and 0.5; tune on val). Optionally add penalty weighting for imputed days or extreme events.

(ii) Direct GraphWaveNet: architecture & equations

Predict E_{T_0} directly without intermediate PM recomposition using an architecture that captures long temporal receptive fields ($L = 30$ days) and learns adaptive adjacency between variables.

Input window length $L = 30$. Stack: Temporal dilated convolution blocks (dilations 1,2,4,8) with residual connections $\rightarrow$ Adaptive graph convolution (learned adjacency using node embeddings) inserted between temporal blocks $\rightarrow$ final MLP readout to predict PET(t).

(a) Temporal dilated conv block

For each node i , the temporal stack applies causal 1D convolutions

with exponentially increasing dilation factors $d \in \{1, 2, 4, 8\}$. Furthermore, kernel size $k = 3$ to obtain a wide receptive field. Residual and skip connections are used to stabilize deeper stacks (Van Den Oord et al., 2016; Wu et al., 2019).

If $T_{:,i}^{(l)}$ is the per-node temporal feature at block l , then:

$$S_{:,i}^{(l)} = \text{Conv1D}_{k=3,d=d_l}(T_{:,i}^{(l)}), \quad (9)$$

with gated activation $\tanh$ and sigmoid gates per WaveNet pattern, residual add, then skip outputs aggregated across layers.

(b) Adaptive graph convolution (learned adjacency)

Learn node embeddings $E \in \mathbb{R}^{N \times p}$. Compute adaptive adjacency:

$$A^{(adapt)} = \text{softmax}(\text{ReLU}(EE^\top)) \quad (10)$$

Optionally combine with static adjacency: $A^* = \beta A^{(static)} + (1 - \beta)A^{(adapt)}$. Apply normalized graph conv $\tilde{A}^* HW$ between temporal blocks to mix variable information using learned relations.

After final temporal + adaptive graph layers, pool node representations (e.g., global readout mapping variable nodes $\rightarrow$ single scalar) using concatenation of selected node embeddings (e.g., radiation node, VPD node), followed by an MLP to predict $\hat{E}_{T_0}$. Optionally predict distributional outputs (mean + log-variance) for heteroscedastic loss.

Primary loss is $\text{MSE}(E_{T_0}^{PM}, \hat{E}_{T_0})$ (when using PM as a surrogate target) or MSE against measurement if in situ PET observations exist. Add L2 regularization and optional quantile/heteroscedastic losses for uncertainty. The complete set of hyperparameters selected for the direct and indirect models is given in Table 2.

2.5. Statistical evaluation criteria

The performance of empirical and hybrid models for PET estimation was assessed using the coefficients of determination (R^2), root mean square error (RMSE), mean absolute error (MAE), and percent bias (PBIAS). Mathematically,

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^N (\text{PET}_{\text{pre}} - \text{PET}_{\text{obs}})^2}{N}}. \quad (11)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\text{PET}_{\text{obs}} - \text{PET}_{\text{pre}}|. \quad (12)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (\text{PET}_{\text{obs}} - \text{PET}_{\text{pre}})^2}{\sum_{i=1}^N (\text{PET}_{\text{obs}} - \text{PET}_{\text{mean}})^2}. \quad (13)$$

$$\text{PBIAS} = \frac{\sum_{i=1}^n (\text{PET}_i^{\text{obs}} - \text{PET}_i^{\text{pre}})}{\sum_{i=1}^n (\text{PET}_i^{\text{obs}})}. \quad (14)$$

Where PET_{obs} , PET_{pre} , and PET_{mean} are observed, predicted, and the mean PET, respectively. The RMSE estimates the mean size of prediction errors and penalizes large deviations, whereas the MAE estimates the mean absolute difference between predictions and observations and is less sensitive to outliers (Waqas and Humphries, 2025). PBIAS indicates the percentage inclination of a model to overestimate or underestimate observed values, with a value of zero indicating no bias. R^2 ranges of 0 to 1 indicate an effective relationship between the predicted and observed values. It is believed that the model is most effective when the R^2 values are close to 1 (Goodwin and Leech, 2006).

3. Results and discussion

The GCRN model was used to establish connections between stations

Table 2
Hyperparameter summary of both direct and indirect DL models.

Parameter	Indirect GCRN	Direct GraphWaveNet
Input window L	14 days	30 days
Node embed dim d	32	32
Graph conv type	Diffusion conv (K = 2)	Adaptive normalized GCN
Graph-GRU layers	2	N/A
GRU hidden units (per node)	128	N/A
Temporal blocks (dilations)	N/A	dilations 1,2,4,8 (kernel=3)
Final readout	Node-wise MLPs → PM	Global MLP → PET
Dropout	0.2	0.2
Learning rate	1e-3	1e-3
Batch size	32	32
Weight decay	1e-5	1e-5
Loss	component MSE + λ_{PM} MSE(PM)	MSE(PET)
$\lambda_{comp}, \lambda_{PM}$	1.0, 0.5 (tune)	N/A
Epochs	≤300 (early stop)	≤300 (early stop)

in the basin (Fig. 3). The graph was plotted with each meteorological station as a node, and edges between stations representing spatial proximity and similarity in the meteorological behavior used to estimate PET. Inter-station distance and climatic similarity measures were used to derive the adjacency structure, enabling the model to capture spatial relationships between stations and then train the GCRN. The 13 meteorological stations are linked by 28 weighted edges that capture the learned spatiotemporal relationships among climatic variables pertinent to PET formation.

The edge weights in this network are the relative strength of interaction exchange between two stations in the GCRN framework. The edge weights are higher, which means that similarity between stations is higher; thus, the meteorological signal of a single station has more influence on the feature aggregation of its neighboring station when performing graph convolution operations. These weights are effective for measuring the strength of co-variation among climatic variables across stations and, hence, for reconstructing PET-related variables in the indirect modeling pathway.

It demonstrates that the network has a moderately high density (0.359), indicating substantial information sharing between stations. The average degree (4.31) suggests that each station makes a significant contribution to the region's representation. The higher edge weights (mean 0.705 ± 0.083) are observed between geographically close or climatically similar stations, indicating the GCRN's ability to capture physically plausible relationships, including common variability in temperature, humidity, and radiation regimes. This trend is related to the anticipated spatial coherence of drivers in the basin, with close stations likely to be more closely related to each other in terms of meteorological forcing and thus have more graph relationships.

There are highly connected nodes (with up to 8°) that serve as regional hubs, where the aggregation of climatic signals is robust. These hub stations are significant to the GCRN because they facilitate the spread of meteorological information across the network, thereby enhancing the model's capacity to capture basin-scale climatic patterns. The high node density also underscores the heterogeneity of drivers the model uses to recreate FAO-56 PM components in the indirect PET estimation pathway.

3.1. Direct and indirect models' evaluation

First, the PET from both models was compared with the FAO-56 PM estimate (Fig. 4a). When higher FAO-56 values (>5.0 mm/day) are required, the GCRN model underestimates PET, as indicated by the regression line trending below the 1:1 reference line. The inaccuracies in the fundamental details (R_n , e_a , VPD) increase when circulated through the PM equation, leading to larger errors at high energies. In contrast,

Graph-WaveNet is more consistent with the 1:1 line in the PET spectrum, with slope values closer to unity and lower dispersion, indicating that the model successfully learns spatiotemporal dependencies with fewer climatic inputs.

Furthermore, Fig. 4b indicates that the Graph-WaveNet has a narrower range of errors, with most deviations clustered around 0.0–0.2 mm, suggesting a decrease in bias and improved generalization. Conversely, the indirect method of the GCRN exhibits greater variability, with station-wise errors reaching ± 3.0 mm, reflecting cumulative errors from the prediction of intermediate components and the recomposition of the FAO-56 PM equation. The kernel density results further support these findings: Graph-WaveNet displays a narrower, more concentrated distribution around zero, while GCRN exhibits a flatter, broader distribution, indicating greater variance and a few instances of underestimation. The predictive accuracy, error symmetry, and stability of Graph-WaveNet trained with the indirect method are superior to those of the direct GCRN framework.

Fig. 5 shows the uncertainty performance of the GCRN and Graph-WaveNet models when predicting PET. The GCRN model has an extremely high probability of coverage of the prediction interval (PICP = 0.994), indicating that most observed values are within the estimated uncertainty limits. Nevertheless, it has a wide mean prediction interval width (MPIW = 7.564), indicating excessively broad and conservative intervals. Graph-WaveNet, in turn, has sufficient coverage (PICP = 0.956) but with much smaller intervals (MPIW = 3.385). It shows a higher reliability-sharpness ratio, and thus it can be concluded that Graph-WaveNet makes more informative and well-calibrated uncertainty predictions for PET.

The comparison is based on statistical evaluation criteria shown in Table 3. GraphWaveNet has the lowest errors, ranging from 0.52 to 0.68 in RMSE and from 0.40 to 0.50 in MAE, rounded to two significant digits, compared to higher errors of 1.3 to 1.6 in RMSE and 1.1 to 1.3 in MAE in the GCRN-direct model. It is a 50–65% decrease in prediction error with GraphWaveNet.

R^2 with GraphWaveNet ranges from 0.85 to 0.92, whereas GCRN ranges from 0.14 to 0.48. These findings suggest that GraphWaveNet accounts for a significantly greater proportion of variance in daily PET, thereby improving the modeling of non-linear meteorological interactions. Moreover, GraphWaveNet exhibits low bias (less than 0.08), whereas GCRN shows a consistent negative bias (0.16–0.69), indicating a systematic underestimation of PET when interactions are not adaptively learned.

Additionally, to validate findings, Taylor diagrams were used, and according to the Taylor diagrams (Fig. 6), in the case of the GCRN model, the observed standard deviation is 1.41–1.6, compared to the predicted standard deviation, which is lower at 1.1–1.3, indicating that temporal PET variability is under-dispersed to a weak extent. Correlations range from 0.75 to 0.90, with stronger correspondence at stations 136 and 278 (0.90), indicating that the GCRN effectively reflects inter-station temporal co-fluctuations. However, it is limited in fully reflecting the variance. This behavior aligns with recurrent architectures, which focus on smoothing over time, eliminating noise, but occasionally softening extremes.

In contrast, GraphWaveNet RMSEs range from 0.66 to 0.89, which correlates with a range of 0.78 to 0.88, indicating that the FAO-56 PM PET can be consistently reconstructed at spatially dispersed stations. The estimated standard deviations (1.02–1.24) are also significantly smaller than the measured standard deviations (1.35–1.61), further indicating that the indirect pathway is more effective at representing the physical variability inherent in radiation and vapor-pressure processes. It is enhanced by (i) the explicit reconstruction of the intermediate components of physics, (ii) the adaptive graph-learning process, which optimizes the relationships between variables beyond the fixed theory of FAO-56.

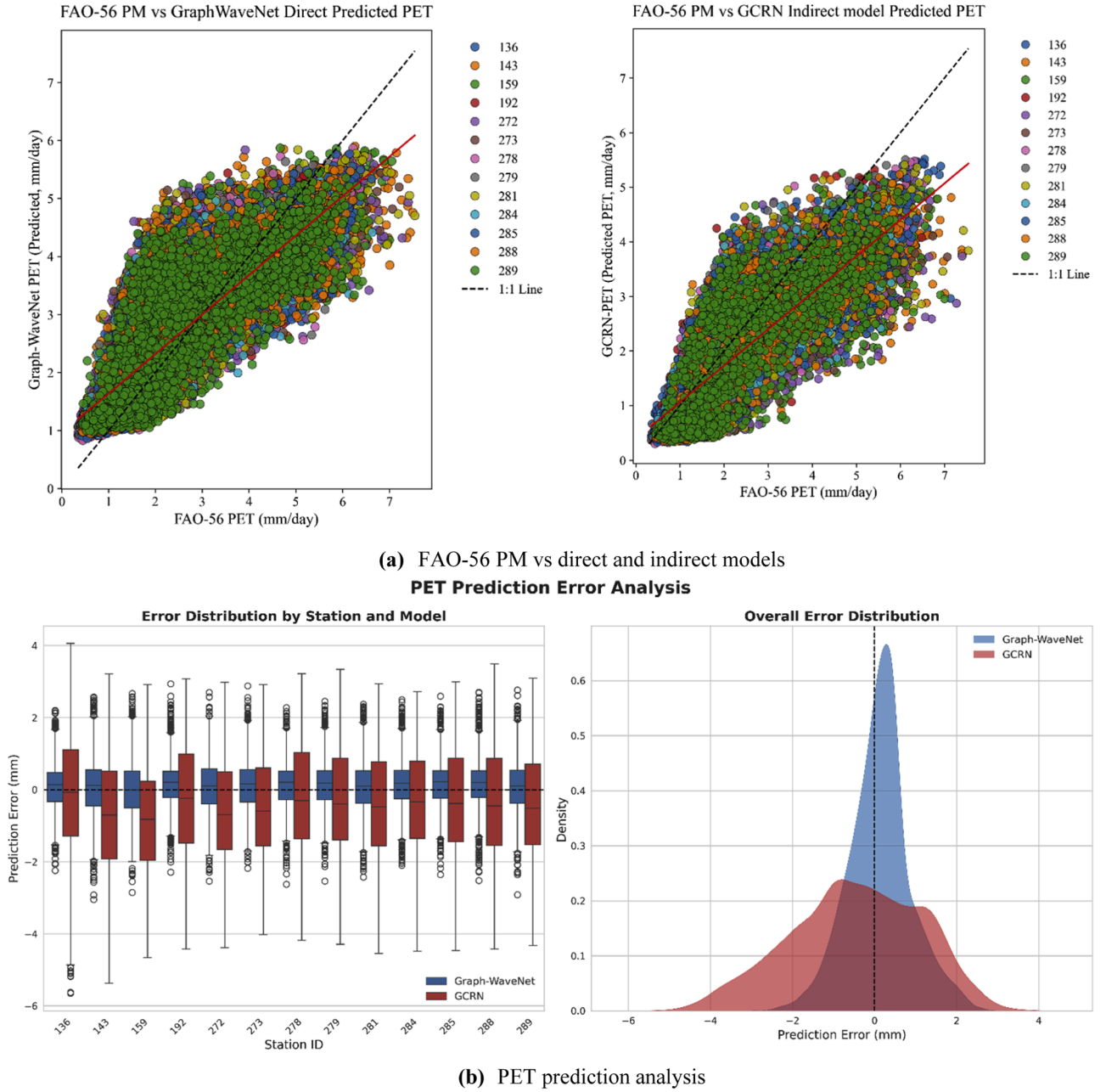


Fig. 4. Direct and model performance assessment: (a) FAO-56 PM vs direct and indirect models, (b) PET prediction error analysis.

3.2. Time series analysis for testing period estimation

The indirect GCRN-PM reconstruction and the direct GraphWaveNet prediction both identify the major seasonal PET dynamics of the basin in the long-term PET time-series analysis (Fig. 7). The low winter daily PET (approximately 1.0–1.5 mm/day) and high summer daily PET (5.0–6.0 mm/day) in the annual PET cycle are repeated in all stations. Correspondence to the FAO-56 PM reference clarifies that the models are learning the energy-driven evapotranspiration regime in the basin.

Nevertheless, the disparities between the indirect and direct approaches are evident in the short-term movements. The results of the GraphWaveNet direct model are more in line with the daily variability, with more accurate peaks and a better response to rapid meteorological variations. GraphWaveNet can learn an adaptive adjacency matrix by modeling variable dependencies, dynamically updating them, and learning non-linear patterns.

The indirect GCRN model produces smoother PET curves than FAO-

56 PM, particularly in summer, when energy levels are high. This smoothing is not surprising, as inaccuracies in the predictions of intermediate components, R_n , e_a , and VPD, spread through the FAO-56 PM equation, resulting in a progressive underestimation of the peak PET. This effect is more severe when FAO-56 PET exceeds approximately 5.0 mm/day, and minor errors in R_n or e_a can result in a large PM numerator. The GCRN's repeated structure is more inclined toward temporal continuity, which suppresses noise and is advantageous for low-variance periods; however, it also suppresses short-period extremes.

In addition to the differences, Fig. 7 shows a significant phase difference between the GCRN-PM estimates and the FAO-56 reference series. The main cause of this phase difference is the sequence of the indirect modeling pathway. Within the GCRN model, meteorological variables are first predicted, then propagated using the FAO-56 PM equation to calculate PET. Any delay in predicting intermediate variables, especially R_n and VPD, which are highly predictive of PET, can shift the timing of the peak PET relative to the reference calculation. The

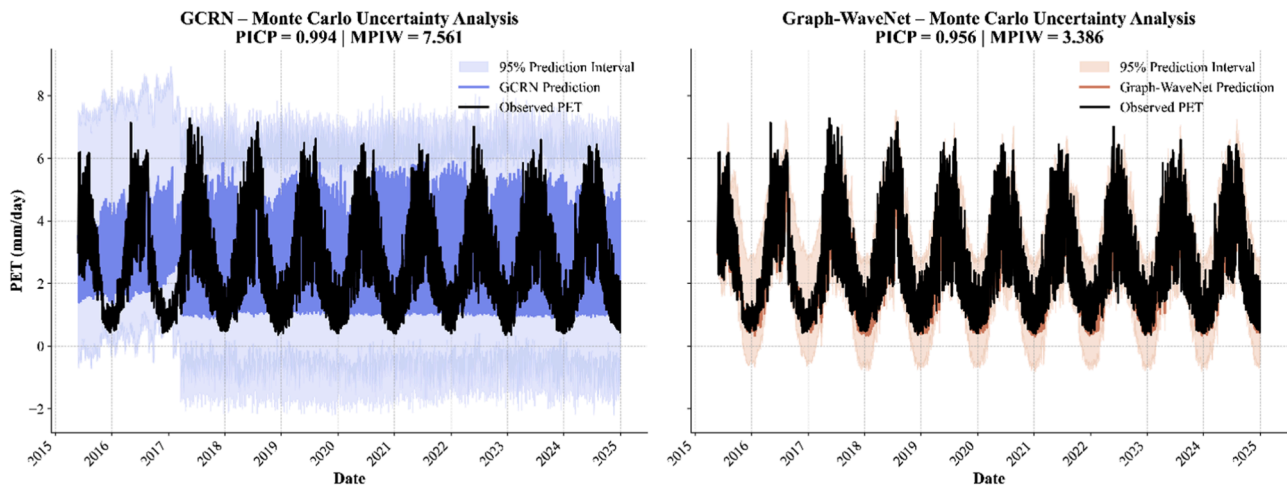


Fig. 5. Comparison of 95% prediction intervals for PET generated by the GCRN and Graph WaveNet using Monte Carlo Dropout-based uncertainty analysis.

repetitive nature of GCRN, though useful for general temporal trends, can also cause minor delays in responding to rapid atmospheric forcing, e.g., sudden surges in solar radiation or humidity variations. Consequently, the predicted PET peaks may be slightly earlier or later than those calculated from the observed meteorological variables.

This difference is one of the reasons why the GCRN-based indirect approach performs comparatively poorly. Even when the magnitude is well represented, error measures such as RMSE and correlation can worsen when the timing of peaks and troughs is shifted relative to the reference series. Hence, the lower performance of the GCRN results is not only due to magnitude smoothing but also to the temporal misalignment introduced during the prediction of intermediate variables and later physical reconstruction.

3.3. Discussion

The results of this study suggest that adaptive spatiotemporal learning integration is more efficient in predicting data and easier to interpret than traditional DL approaches. This tendency is consistent with the growing evidence that hybrid methods, combining domain knowledge and data-driven learning, are more effective than purely empirical methods of hydroclimatic prediction (Ferreira et al., 2022; Gyamfi et al.; Stefanidis et al., 2025; Wi and Steinschneider, 2024).

The current study identified relational dependencies among meteorological variables, unlike traditional MLP, CNN, or LSTM models, which assume that feature types are independent of one another (Stefanidis et al., 2025; Waqas et al., 2023; Waqas and Wangwongchai, 2025; Wu et al., 2020). Similar to other physics-informed graph DL models (Jia et al., 2023; Zhao et al., 2021, 2024), the indirect GCRN model can model regular interactions among radiation, humidity, vapor pressure, and temperature and encode them in a learned graph topology. Representations formed latently replicated the FAO-56-PM components with moderate fidelity at low- to mid-range PET conditions. It confirms previous findings that hybrid DL-process models are easier to interpret and physically realistic (Sun et al., 2024; Waqas and Wangwongchai, 2025; Wi and Steinschneider, 2024).

Nonetheless, the indirect GCRN showed higher errors at minimum PET concentrations (greater than 5 mm/day), primarily due to error amplification by the PM equation. The same results have been observed in component-wise ET modeling, where small errors in estimating radiation or vapor pressure result in significant errors in the PM numerator (Allen et al., 2011; Almorox et al., 2018). Extreme PETs are also smoothed in recurrent networks (Hilmi et al., 2022), which explains why the effect of systematic underestimation was observed in summer peaks. This weakness highlights the challenge of reconstructing complex

energy balance processes when learned subcomponents are involved, and the received variables do not respond linearly to climate variability.

Conversely, direct GraphWaveNet achieved higher accuracy, greater variance consistency, and greater bias reduction across all stations. This fact (a 20- or 30-point lower RMSE) shows the strength of its dual-receptive-field design, which combines dilated temporal convolutions with adaptive adjacency learning. GraphWaveNet could also learn long-range temporal dependencies, including those not mediated by intermediate variables (Van Den Oord et al., 2016; Wu et al., 2019), and dynamically learn inter-variable relationships from data-driven similarity patterns, which was better than the fixed physical graph employed in the GCRN. It is consistent with the recent body of knowledge in hydrometeorological DL, which shows that adaptive graph structures are associated with greater robustness and scalability, and that such architectures perform better in heterogeneous environments (Eder, 2024; Zhao et al., 2021).

The good performance of the direct model also indicates that PET estimated well even in the presence of missing critical physical inputs. This result aligns with the current literature, which indicates that DL models trained on limited meteorological data can still provide strong ET estimates (Ferreira et al., 2022; Waqas and Kim, 2025; Yan et al., 2023). The demonstrated capability of the GraphWaveNet to operate in data-sparse conditions reinforces its ability to support real-time water resource management, drought detection, and climate-impact forecasting.

Furthermore, both models accurately explained the seasonal dynamics of PET within the basin, supporting the notion that they can encode regional climatological regimes characterized by monsoonal radiation and temperature trends. PET modeling in Korea and in other complex-terrain basins worldwide has also reported similar spatially dependent DL performance (Xiong et al., 2024). These results also confirm that adaptive graph structures are essential for localized non-linearities, as evidenced by GraphWaveNet's higher-quality performance in these areas.

Irrespective of the merits, the study has some weaknesses: (1) the indirect pathway is sensitive to errors that can accumulate during the reconstruction of intermediate variables. Although this design is more straightforward to interpret, it might require more sophisticated physics-informed constraints or multi-task learning models, as implied by current studies on hybrid ET (Slater et al., 2023; Sun et al., 2024). (2) The models were based on the FAO-56 PM outputs as ground truth, i.e., errors in PM spreading were included in the training, especially during humid or monsoon seasons when PM assumptions might not be entirely valid (Okkan et al., 2024). (3) Despite GraphWaveNet's good performance, its adaptive adjacency mechanism can be prone to overfitting

Table 3

DL model's evaluations based on metrics.

Station ID	GraphWaveNet direct model				GCRN-direct model			
	RMSE	MAE	R ²	Bias	RMSE	MAE	R ²	Bias
288	0.60	0.45	0.88	0.05	1.45	1.18	0.44	-0.29
289	0.63	0.47	0.87	0.03	1.41	1.14	0.42	-0.31
192	0.58	0.44	0.89	0.05	1.42	1.16	0.45	-0.19
136	0.52	0.40	0.92	0.02	1.63	1.28	0.14	-0.16
143	0.68	0.50	0.86	0.01	1.57	1.28	0.38	-0.55
272	0.62	0.46	0.88	0.04	1.38	1.12	0.41	-0.47
273	0.59	0.44	0.89	0.06	1.34	1.09	0.46	-0.39
278	0.54	0.42	0.91	0.04	1.36	1.11	0.48	-0.21
279	0.57	0.43	0.90	0.05	1.41	1.15	0.44	-0.26
281	0.61	0.45	0.88	0.04	1.40	1.14	0.39	-0.33
284	0.58	0.43	0.90	0.08	1.32	1.08	0.47	-0.28
285	0.56	0.42	0.90	0.05	1.35	1.10	0.43	-0.27
159	0.67	0.50	0.85	0.02	1.57	1.26	0.22	-0.69

without regularization, particularly in small basins or short datasets. (4) It is not clear whether this research evaluated generalizability across basins; transfer learning or cross-basin validation could provide additional insights into model performance.

The results indicate that physical interpretability, coupled with adaptive deep spatiotemporal modeling, is a potent and scalable solution for PET estimation. The complementary nature of indirect and direct suggests that future studies can leverage hybrid networks that combine physics-based reconstruction and adaptive graph learning within a single architecture.

For future studies, recent findings indicate that transfer learning approaches can significantly improve model generalization across sites with diverse climatic and topographic conditions, thereby increasing the transferability of data-driven PET models (Amanatidis et al., 2026). Recent research also shows that bi-directional transfer learning between Mediterranean forest ecosystems along elevation gradients can achieve strong predictive performance, even with limited data (Amanatidis et al., 2026). Therefore, the use of transfer learning approaches can potentially enhance PET predictions in various hydroclimatic zones of Korea.

4. Conclusion

This study investigated the performance of graph-based DL models for estimating daily PET in the Nakdong River basin using daily meteorological observations and FAO-56 PM-based PET values. Two DL modeling strategies were explored: an indirect, physics-informed

strategy based on a GCRN, which forecasts intermediate physical variables before PET reconstruction, and a direct GraphWaveNet architecture that predicts PET directly as a function of meteorological inputs.

The results of quantitative assessment at 13 meteorological stations showed that the direct GraphWaveNet model significantly surpassed the indirect GCRN framework. GraphWaveNet had much smaller prediction errors (RMSE: 0.52–0.68 mm day⁻¹; MAE: 0.40–0.50 mm day⁻¹) and higher explanatory power (R²: 0.85–0.92), which means that it is better able to capture non-linear spatiotemporal interactions between meteorological variables. Conversely, the indirect GCRN method systematically underestimates in high-PET conditions due to cumulative errors in predicting intermediate variables and their propagation through the PM equation.

The results show that adaptive spatiotemporal graph learning provides an effective and powerful framework for PET prediction, especially in data-limited settings. Graph neural networks have the potential to be applied to more complex hydrometeorological modeling and water resource management tasks, as incorporating relational dependencies among hydroclimatic variables improves model generalization and predictive accuracy.

List of abbreviations

PET	Potential Evapotranspiration
ET	Evapotranspiration
PM	Penman–Monteith
FAO	Food and Agriculture Organization
DL	Deep Learning
ML	Machine Learning
GNN	Graph Neural Network
GCRN	Graph Convolutional Recurrent Network
GRU	Gated Recurrent Unit
MLP	Multilayer Perceptron
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
R ²	Coefficient of Determination
N	Sunshine Duration
T _{min}	Minimum Temperature
T _{max}	Maximum Temperature
R _H	Relative Humidity
U _z	Wind Speed at 2 m Height
R _n	Net Radiation
VPD	Vapor Pressure Deficit
e _a	Actual Vapor Pressure
e _s	Saturation Vapor Pressure
MPIW	Mean Prediction Interval Width

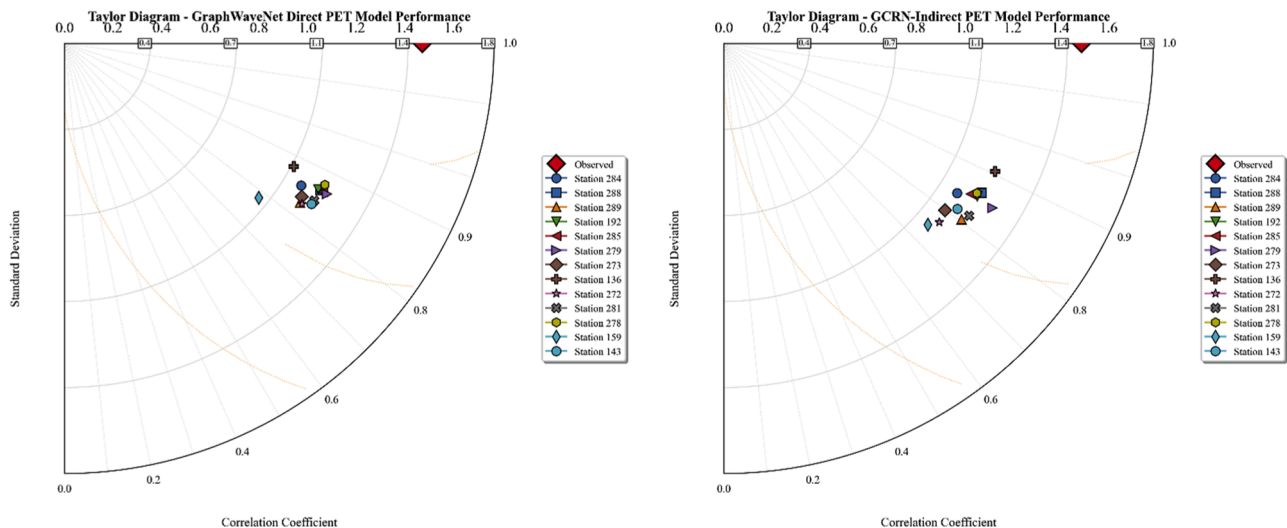


Fig. 6. Taylor diagrams comparing the GCRN model and the GraphWaveNet model with the FAO-56 PM method as the reference.

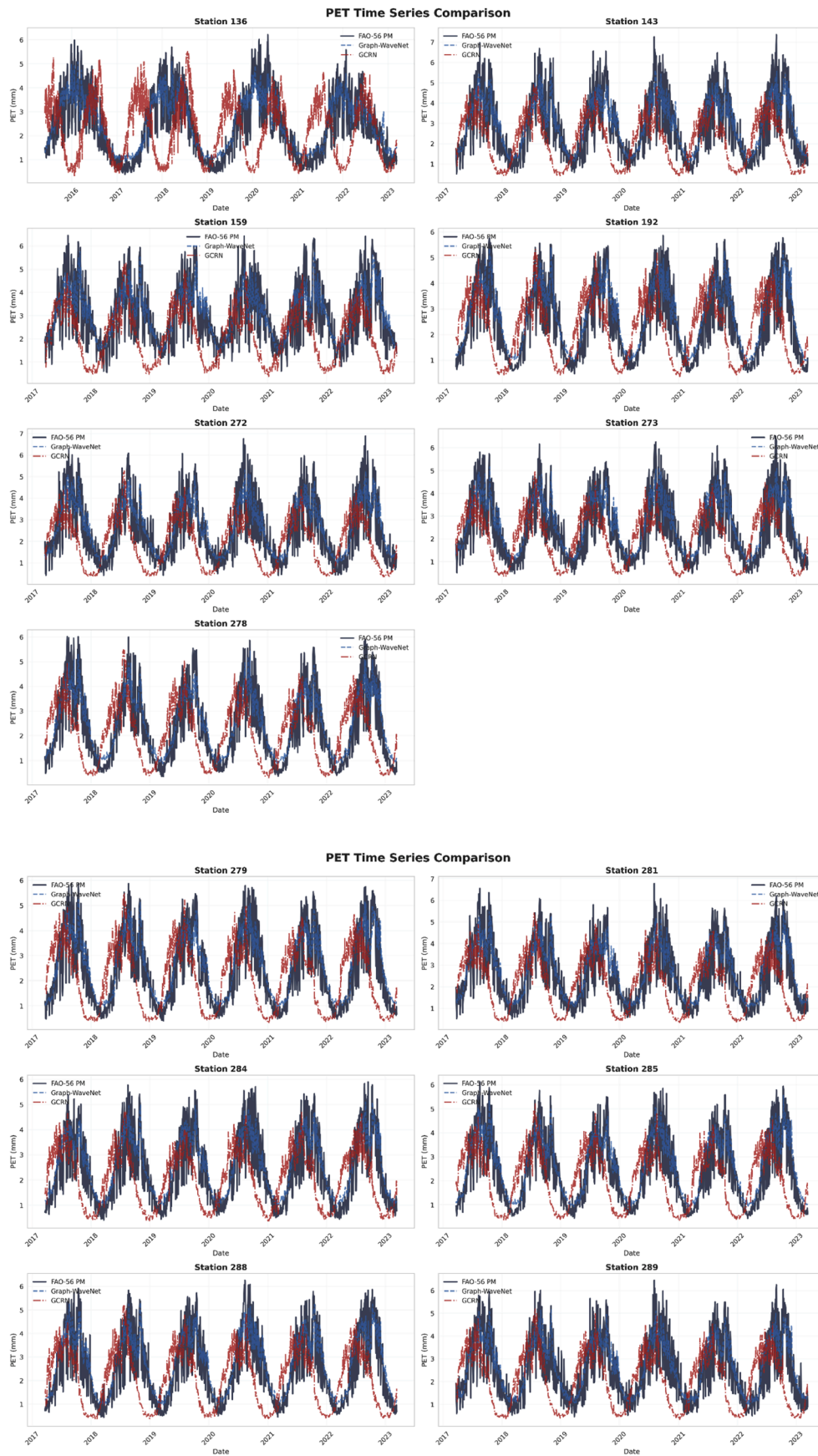


Fig. 7. Station-wise comparison of PET estimation using direct and indirect models.

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Data availability statement

Data will be available on request from the corresponding author.

CRedit authorship contribution statement

Muhammad Waqas: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Sang Min Kim:** Supervision, Resources, Project administration, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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